Answers on
Booklet - 8

Taken from OCR Physics A past papers

Abbreviations, annotations and conventions used in the Mark Scheme	/	=	alternative and acceptable answers for the same marking point
	;	=	separates marking points
	NOT	=	answers which are not worthy of credit
	( )	=	words which are not essential to gain credit
	____	=	(underlining) key words which **must** be used to gain credit
	ecf	=	error carried forward

1.

cast iron:	brittle brittle explained as having no plastic region elastic elastic explained as returning to original length when the load is removed / linear graph / Hooke's law obeyed or equivalent words
	MAX 3
copper:	ductile ductile explained as can be formed into a wire initially elastic plastic where it stretches more and more with little increase in stress plastic explained as does not return to its original length when the load is removed reference to necking at the end
	MAX 3
polythene:	easy to deform / deformed with a small force plastic ductile polymeric
	MAX 2

MAX 8

2.

(a)(i)	extension / original length or unit length	B1
(ii)	force / (cross – sectional) area	B1
(b)	Pa / $N\ m^{-2}$	B1
(c)	measurements:	
	area / diameter, original length, force / mass and extension	B1
	micrometer for the diameter	B1
	(metre) rule for the original length	B1
	masses used for the load / force = mass x g	B1
	marker on the wire and mm scale / rule for extension	B1
		[MAX 3]
	length of wire greater than one metre stated	B1
	repeating for a number of loads	B1
	measure the diameter in several places	B1
	measure the original length to the marker	B1
	marker put nearer to pulley	B1
		[MAX 3]
	allow one mark for good physics	B1
	eg using a second wire as a reference or to compensate for temperature changes / use a second wire with a micrometer scale to measure the extension / remove load to check extension	
	calculations:	
	plot a graph (eg force against extension of equivalent)	B1
	determine the gradient	B1
	E = (gradient x length) / area	B1
	substituting values into the equation scores one out of three	

3.

(a)	**(E)** = stress/strain	**B1**
(b)(i)	strain = extension/length	**C1**
	= $4.2 \times 10^{-3} / 0.75$	
	= 5.6×10^{-3}	**A1**
(b)(ii)	E = force / (area x strain)	**C1**
	F = E x A x strain	
	= $2 \times 10^{11} \times 4.5 \times 10^{-7} \times 5.6 \times 10^{-3}$	**C1**
	= 504 (N)	**A1**
(c)	larger extension in words scores 1	
	(twice the extension) 8.4 scores 2	**B2**
(d)(i)	density = mass / volume	**B1**
(ii)	volume larger hence density less	**B1**

4.

i	$\rho = m/V = m/Av$; $m = A\rho v = 7.5 \times 10^{-5} \times 1000 \times v = 0.09$ giving $v = 1.2$ m s^{-1}
ii	2.4 (m s^{-1})
iii	F = d (mv)/dt /AW; F = 0.09 x (2.4 – 1.2) ;= 0.11 (N) ; *ecf (a)ii*
iv	towards or into shower head/backwards *ecf (a)iii*

5.

a	**i**	48 (N) 0.25 (s)	**1**
	ii	estimating area under graph or mean F; 6.5 ± 1	**2**
b	**i**	a = F/m or = 48/0.5 ; = 96 (m s^{-2}) *ecf a(i)*	**2**
	ii	Ft = mv; v = *a(ii)*/0.5 = 2*a(ii)* (m s^{-1}) *ecf a(ii)*	**2**
	iii	k.e. = ½ mv^2 or = ½ x 0.5 x *b(ii)*^{2} ; = *a(ii)*^{2} (J) *ecf b(ii)*	**2**
c		Ft = mv ± mu or = 0.5 (8 ± 14) ;	**1**
		F = 11/0.18 ; = 61(.1) (N) *aliter mean a = 12(2) m s^{-2} F = ma*	**2**

6.

Resultant force (and resultant torque) is zero
Weight (down) = force from/due to spring (up)

0.2, 0.6, 1.0 s (*one of these*)

0, 0.8 s (*one of these*)

0.2, 0.6, 1.0 s (*one of these*)

Hooke's law: extension is proportional to the force (*not mass*)
Linear/straight line graph hence obeys Hooke's law

Use of the gradient (*not just* $F = kx$)
$K = (0.4 \times 9.8) / 15 \times 10^{-2}$
$= 26(.1)\ \mathrm{N\,m^{-1}}$

either energy = area to left of line *or* energy = $\frac{1}{2} ke^2$
$= \frac{1}{2} \times [(0.4 \times 9.8) / 15 \times 10^{-2}] \times (15 \times 10^{-2})^2$
$= 0.294\ \mathrm{J}$ (*allow 2 s.f.*)

7.

area between lines represents energy/area under curve represents energy when rubber is stretched and then released/two areas are different
this energy seen as thermal energy/heating/difference represents energy released as heat

8.

kinetic energy $= \frac{1}{2} mv^2$
$= 0.5 \times 600\ (9.5)^2$
$= 27075\ (\mathrm{J}) = 27\ \mathrm{kJ}$

potential energy $= mgh$
$= 600 \times 9.81 \times 4.1$
$= 24132\ (\mathrm{J})$
$= 24\ \mathrm{kJ}$

work done = 27 – 24 = 3.0 kJ

resistive force = 3000 / 8.2 (distance along slope = 4.1 / sin 30°)
= 366 N

9.

(a)	*Longitudinal* any valid example e.g. sound /ultra sound (accept 'P waves' but reject "Earthquake") *Transverse* any valid example e.g. light / microwaves etc./ emwaves / any emradiation (accept sea waves or ripple-tank waves but not "water waves")	 B1 B1	 [2]
(b)(i)	frequency = number of waves/vibrations/oscillations/cycles per unit time {allow "per second"}	B1	[1]
(ii)	period = time taken for 1 complete cycle/wave/vibration/oscillation {allow 1/f, provided f has been correctly defined above}	B1	[1]
(c)(i)	4×10^{-5} m { not 4 or 4×10^{5}}	B1	[1]
(ii)	air particle vibrates (seen or implied in remainder of answer) in the same direction as the wave (WTTE) {allow left to right, backwards and forwards, longitudinally but NOT up and down} description of motion during any full cycle (allow ecf for transverse) *{do not penalise reference to random thermal motion of air molecules}*	B1 B1 B1	 [3]
(iii)	***similarity***: e.g. both have same amplitude / frequency / period OR move/vibrate/travel longitudinally OR both have (random) thermal motion ***difference:*** e.g. movements (90°) out of phase (WTTE)	 B1 B1	 [2]
(iv)	wavelength = 0.8 m	B1	[1]
(v)	recall of $v = f\lambda$ correct substitution: $f = v/\lambda = 340/0.8$ f = 425 Hz	C1 C1 A1	 [3]

10.

Maximum of 2 marks for correctly identifying the 4 errors OR stating the 2 correct notes:
i.e. errors in notes 1, 2, 3, and 6 (shown anywhere) B2
{5 or 6 or 2 or 1 notes nominated scores ZERO, 4 correct scores 2, 3 correct scores 1}

Valid corrections score 1 mark each: do not allow "NOT" corrections apart from note 3
Note 1: In longitudinal waves vibrations are parallel to wave direction (WTTE) B1
{OR in transverse waves vibrations are perpendicular to wave direction (WTTE}

Note 2 light (or any of the em waves) can travel through a vacuum (WTTE) B1
{ allow sound/longitudinal waves cannot travel thro' a vacuum}

Note 3: waves carry energy/disturbance (not displacement or info) from..... (WTTE) B1
{allow "waves do not carry the medium" and "the medium carries the waves from....." }

Note 6: wavelength = distance from crest to crest/trough to trough/max to max ((WTTE) B1

11.

(a) Flow / movement of charge / charged particle(s) AW B1
(Allow current = rate of flow of charge / current = rate of change of charge)

(b) The charge (flowing past a point) in 1 s when current is 1 A B1
(Allow 1C = 1A × 1s)

(c)(i) $I = \frac{\Delta Q}{\Delta t}$ / $I = \frac{340}{50}$ (Allow any subject - with or without Δ) C1
6.8 (A) A1

12.

a		Circle round C s^{-1}	B1	
b		Correct arrows at O and P – 'clockwise'	B1	**Allow** a single arrow.
c		$I = Q/t$ / charge = current × time	C1	The first mark is for a correct equation – any subject
		$Q = 9.4 \times 10^{20} \times 1.6 \times 10^{-19}$ (= 150.4 C)	C1	
		$I = \frac{150.4}{30}$	C1	There are 3 marks for correct values of charge and time substituted into the equation – current must be the subject
		current = 5.0(1) (A)	A0	**Reject** A bald answer of 5 A (since this is given in the question) **Allow** 1 mark for a bald answer of 5.01 (A) – 3 sf or more.